



IPB University
— Bogor Indonesia —

Departemen
Ilmu Komputer

Kecerdasan Buatan (*Artificial Intelligence*) KOM321

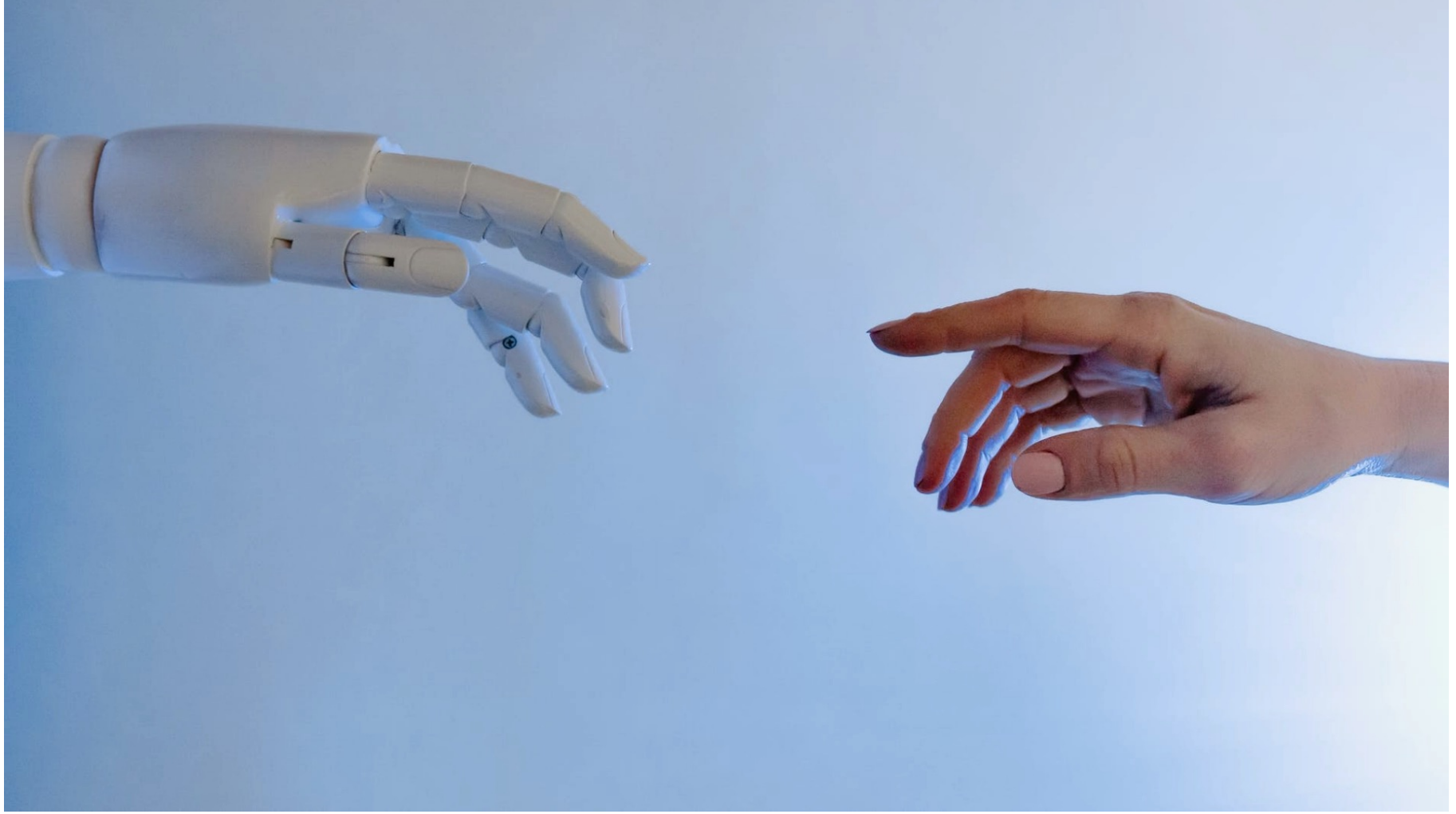
Tim Pengajar :

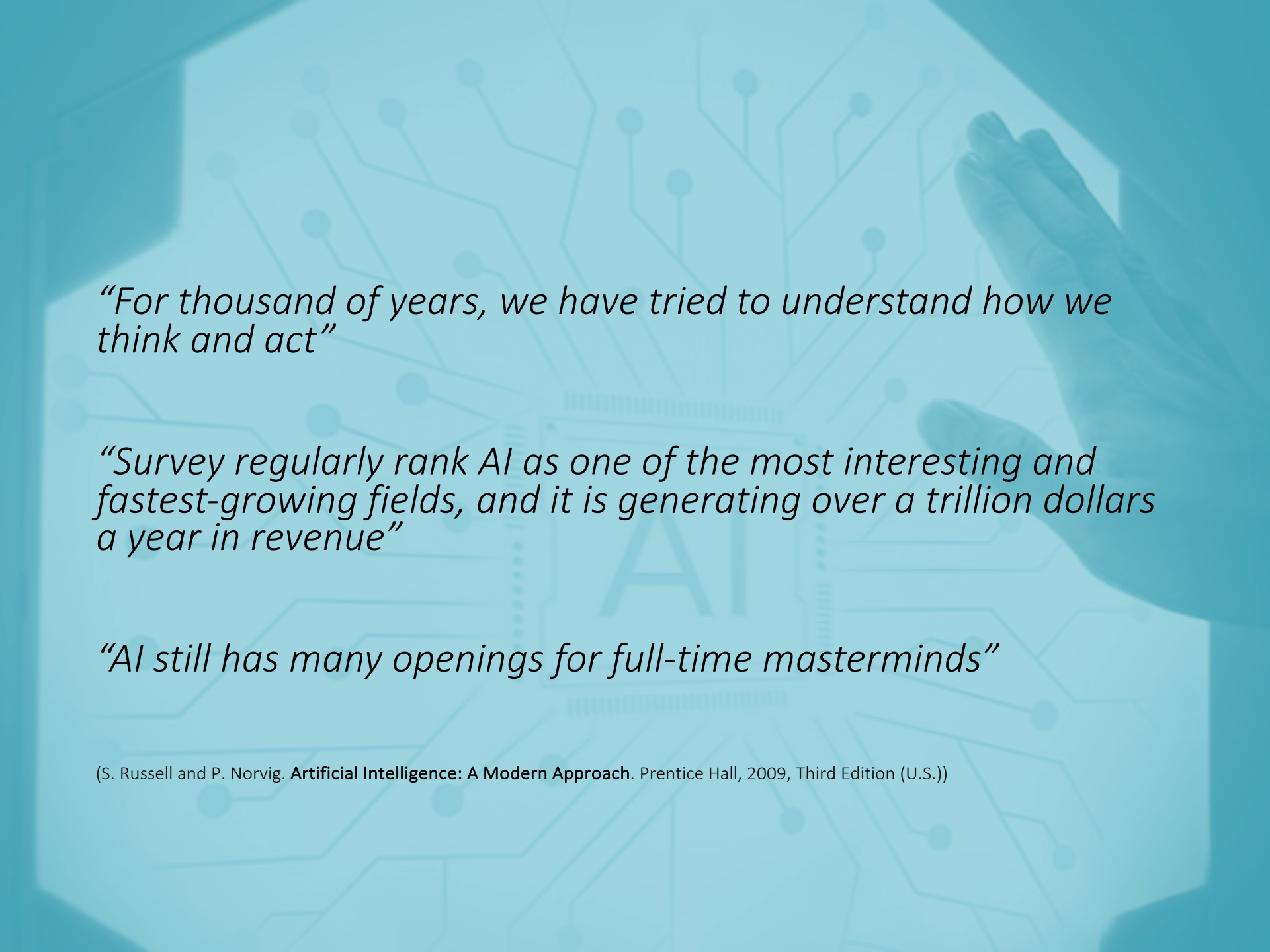
Dr. Yeni Herdiyeni (YHY) (Koordinator)

Dr. Medria Hardhienata (MKD)

Dr. Karlisa Priandana (KPR)

Dr Sony Hartono Wijaya, SKom, MKom (SHW)



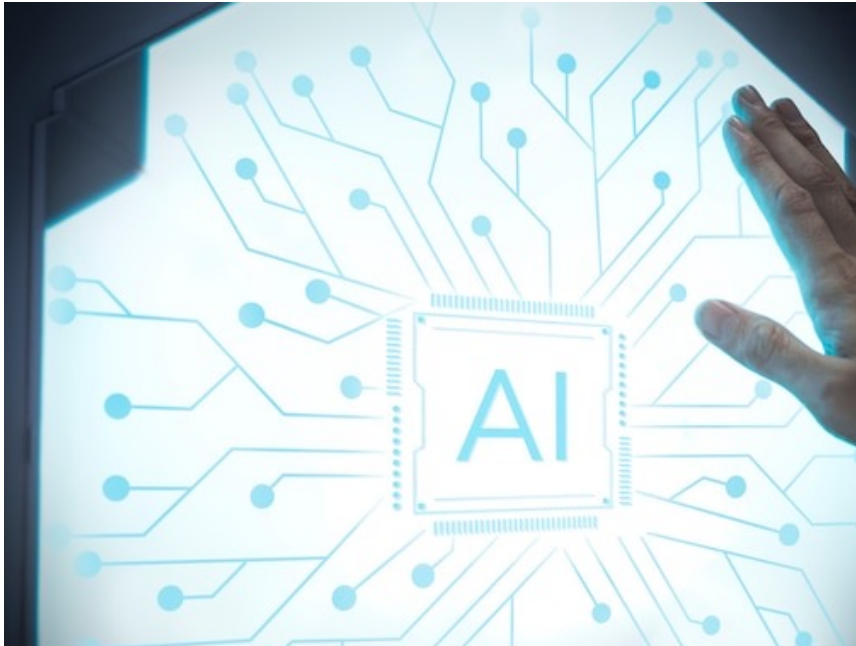
The background is a light blue gradient with a faint, stylized circuit board pattern. A large, semi-transparent hand is visible on the right side, reaching towards the center. The text is in a black, italicized serif font.

“For thousand of years, we have tried to understand how we think and act”

“Survey regularly rank AI as one of the most interesting and fastest-growing fields, and it is generating over a trillion dollars a year in revenue”

“AI still has many openings for full-time masterminds”

(S. Russell and P. Norvig. **Artificial Intelligence: A Modern Approach**. Prentice Hall, 2009, Third Edition (U.S.))



Kecerdasan Buatan (KOM321)

Kuliah:

Rabu, 13.00 – 14.40 (K1)

Selasa 10.00 – 11.40 (K2)

Selasa 13.00 – 14.40 (K3)

Praktikum:

Kamis, 07.00 - 09.00 (P1)

Kamis, 07.00 - 09.00 (P2)

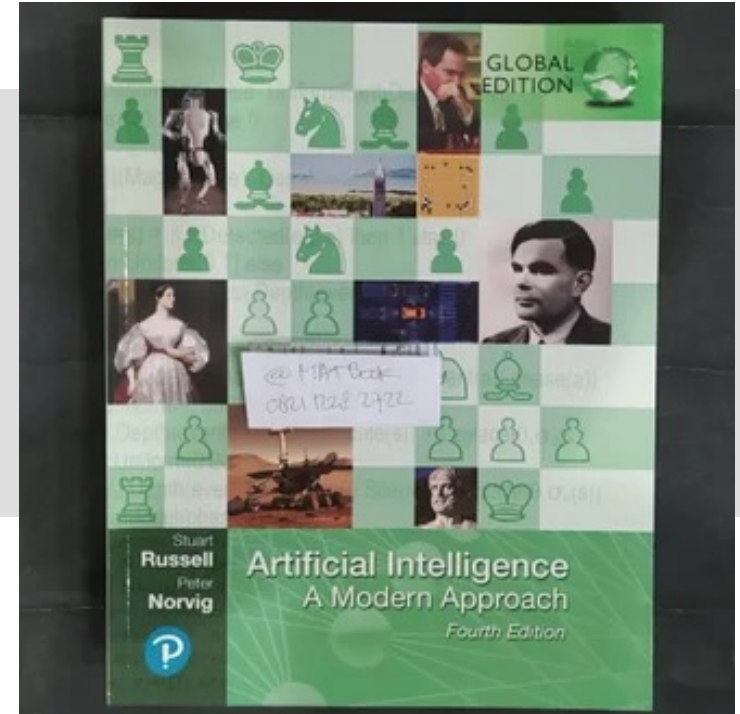
Kamis, 15.30 – 17.30 (P3)

LMS :

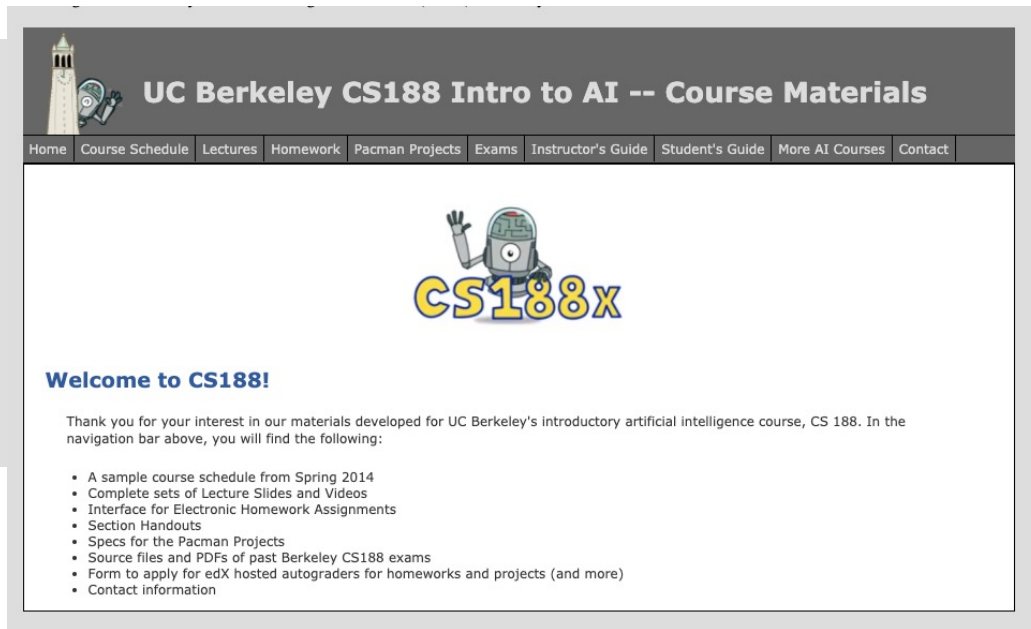
[KOM321 Kecerdasan Buatan](#)

Buku Referensi Utama

Buku: S. Russell and P. Norvig.
Artificial Intelligence: A Modern Approach. Prentice Hall, 2009, Third Edition (U.S.)



Referensi Lain



<http://ai.berkeley.edu/home.html>

Praktikum

Bahasa Pemrograman: Python 3.4/3.5



Topik perkuliahan

Topik	Subtopik	Jumlah Pertemuan
Pendahuluan (Chap 1)	Penjelasan silabus, pengenalan kecerdasan buatan, pengenalan python	1 pertemuan
Implementasi Agen dan Environment (Chap 2)	Penjelasan agent dan environment dalam kecerdasan buatan	1 pertemuan
Penelusuran/searching (Chap 3-4)	Informed search	1. pertemuan
	Uninformed search	1 pertemuan
Adversarial search (Chap 5)	CSP, Games, or Adversarial Search (map coloring problem, n-queen problems, zebra-puzzle)	1 pertemuan
Knowledge Representation and Inference for logic (proportional logic) (Chap 7-9)	Proportional Logic	1 pertemuan
	First Order Logic	1 pertemuan
UTS		

Topik perkuliahan

Topik	Subtopik	Jumlah Pertemuan
Inference Techniques (Chap 10)	Inference (forward, backward, resolution)	1 pertemuan
Statistical Inference (Chap 13-15)	Statistical Learning : Naive Bayes	1 pertemuan
Statistical Inference (Chap 13-15)	Statistical Learning: Bayesian Network	1 pertemuan
Learn to estimate functions from examples (Chap 18-20)	Learning from Observation: Nearest Neighbor, Decision Tree	1 pertemuan
Neural Network	Neural Network, Deep Learning	1 pertemuan
Deep Learning	Swarm Intelligence, Evolutionary Computing	1 pertemuan
Project	Student Project	1 Pertemuan
UAS		

Rencana Perkuliahan

		Pertemuan Ke-													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14
K2	Pengajar	MKD	MKD	MKD	MKD	KPR	KPR	KPR	KPR	YHY	YHY	YHY	WUL	WUL	WUL
	Daring/ Luring	Daring	Luring	Daring	Luring	Luring	Daring	Daring	Luring	Daring	Daring	Daring	Luring	Luring	Luring
K3	Pengajar	MKD	MKD	MKD	MKD	KPR	KPR	KPR	KPR	AZK	AZK	AZK	AZK	MKD	MKD
	Daring/ Luring	Daring	Luring	Daring	Luring	Luring	Daring	Daring	Luring	Luring	Luring	Daring	Daring	Daring	Luring
K1	Pengajar	YHY	YHY	YHY	YHY	YHY	YHY	YHY	YHY	SHW	SHW	SHW	WUL	WUL	WUL
	Daring /Luring	Daring	Luring	Luring	Luring	Luring	Daring	Daring	Luring	Daring	Daring	Luring	Daring	Daring	Luring

K1: Kelas Paralel 3, Jadwal Kuliah Rabu pukul 13.00-14.40 WIB

K2: Kelas Paralel 1, Jadwal Kuliah Selasa pukul 10.00-11.40 WIB

K3: Kelas Paralel 2, Jadwal Kuliah Selasa pukul 13.00-14.40 WIB

Penilaian

Nilai akhir (NA) adalah nilai kumulatif dari:

- Nilai ujian tengah semester (UTS)
- Ujian akhir semester (UAS)
- Tugas perorangan (TP)
- Tugas kelompok atau proyek akhir (PA)

Metode dan bobot nilai sebagai berikut:

UTS (1-7) dan UAS (8-14) dilakukan melalui ujian UTS dan UAS dengan bobot masing-masing 30%.

Nilai Praktikum diberi bobot 10%

Nilai TP adalah rata-rata dari semua tugas yang diberikan, dan diberi bobot 10%

Nilai PA terdiri dari nilai produk proyek: program komputer, laporan dan presentasi.

Bobot nilai PA adalah 20%.

Penilaian

Selang nilai untuk menetapkan huruf mutu A, AB, B, BC, C, D, atau E ditentukan berdasarkan nilai rata-rata dan simpangan baku dengan menggunakan sebaran normal serta pertimbangan lainnya.



Absensi

Mahasiswa diwajibkan menghadiri kuliah setidaknya 80% dari seluruh pertemuan. Bagi mahasiswa yang kehadirannya tidak mencapai 80%, maka mahasiswa tersebut tidak diperbolehkan mengikuti ujian akhir semester, sehingga nilai ujian akhir semesternya dinyatakan 0.



Absensi

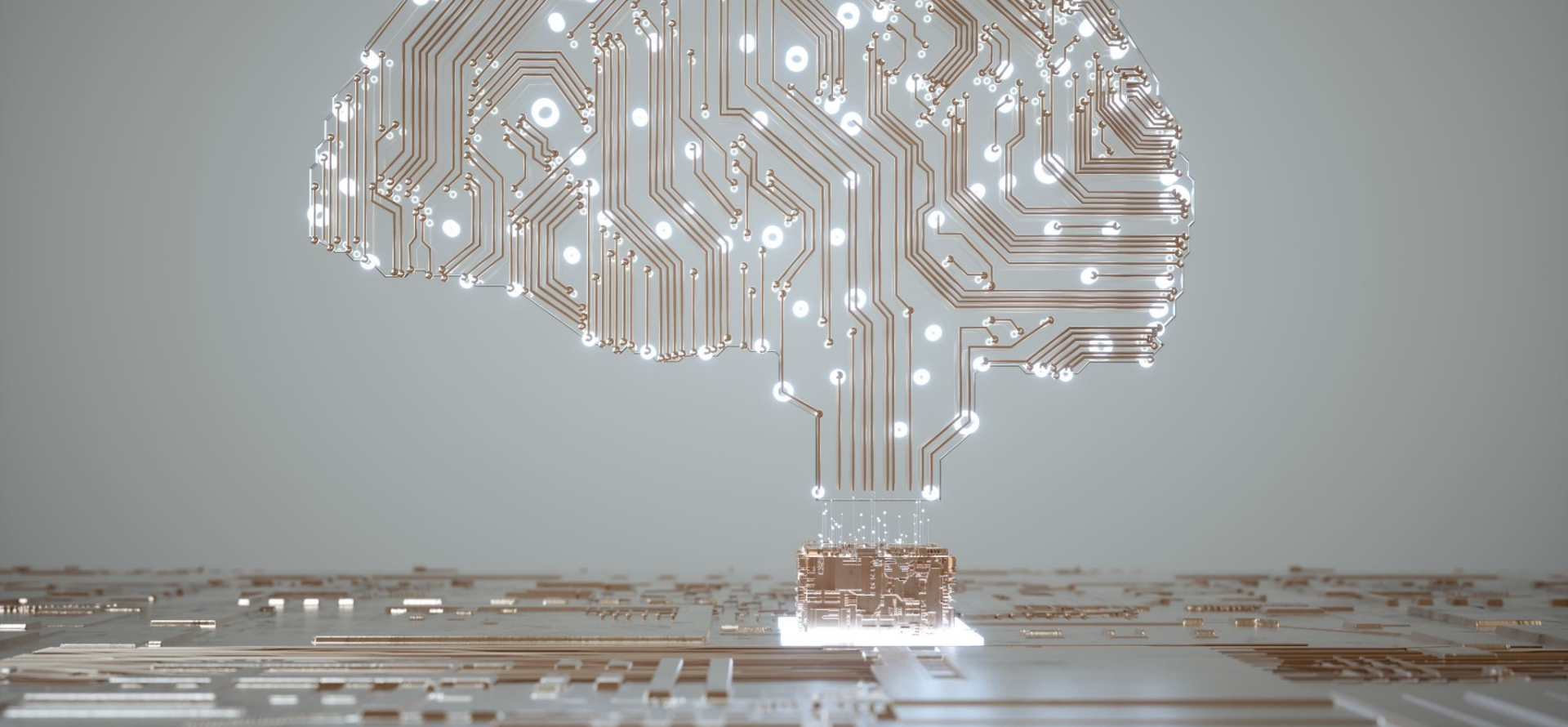
Setiap mahasiswa diwajibkan mengikuti seluruh kegiatan praktikum (kehadiran praktikum: 100%). Jika mahasiswa tidak mengikuti praktikum maka komponen tugas pada praktikum yang tidak dihadiri adalah 0 dan kepada mahasiswa tersebut tidak diberikan tugas praktikum susulan.

Tugas praktikum susulan hanya diberikan kepada mahasiswa yang tidak hadir dalam praktikum dikarenakan sakit yang ditunjukkan dengan surat keterangan sakit dari dokter.



Pengumuman

Pengumuman nama-nama mahasiswa yang tidak dapat mengikuti ujian akhir semester akan diberikan selambat-lambatnya 3 (tiga) hari sebelum ujian akhir semester dilaksanakan.



Pengenalan Kecerdasan Buatan (*Artificial Intelligence - AI*)

Kuliah 01



Apa definisi Kecerdasan Buatan
(*Artificial Intelligence*)?

Let's play a game!

Join at menti.com use code 84 31 90 7

Mentimeter

Apa yang dimaksud dengan Kecerdasan Buatan (What is Artificial intelligence)?



0	0	0	0
Putting your intelligence into Computer	Programming with your own intelligence	Making a Machine intelligent	Playing a Game



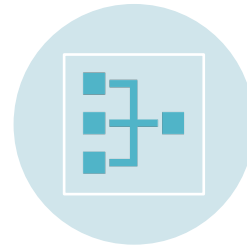
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Artificial Intelligence (AI)



Otomatisasi kegiatan yang dikaitkan dengan pemikiran manusia, seperti:

kegiatan pengambilan keputusan (*decision-making*), pemecahan masalah (*problem-solving*), dan proses belajar (*learning*)..." (Bellman, 1978)



"Seni menciptakan mesin yang dapat mengerjakan fungsi/tugas yang membutuhkan kecerdasan jika dilakukan oleh manusia" (Kurzweil, 1990)

“Sebuah upaya baru yang menarik untuk membuat komputer berpikir... mesin yang dilengkapi dengan akal pikiran (memiliki kemampuan berpikir), dalam arti sepenuhnya dan makna sesungguhnya” (Haugeland, 1985) .

“Otomatisasi kegiatan yang dikaitkan dengan pemikiran manusia, seperti kegiatan pengambilan keputusan (*decision-making*), pemecahan masalah (*problem-solving*), dan proses belajar (*learning*)...” (Bellman, 1978).

“Sebuah studi tentang kemampuan mental melalui penggunaan model komputasi” (Charniak and McDermott, 1985).

“Studi komputasi yang memungkinkan terbentuknya persepsi, penalaran, dan tindakan” (Winston, 1992)

“Seni menciptakan mesin yang dapat mengerjakan fungsi/tugas yang membutuhkan kecerdasan jika dilakukan oleh manusia” (Kurzweil, 1990)

“Sebuah studi tentang bagaimana membuat komputer melakukan sesuatu dimana, pada saat ini, manusia melakukannya dengan lebih baik”- (Rich dan Knight, 1991)

“Sebuah bidang studi yang bertujuan untuk menjelaskan dan meniru perilaku cerdas dalam hal proses komputasi” (Schalkoff, 1990)

“Cabang dari Ilmu Komputer yang berkaitan dengan otomatisasi perilaku cerdas (*intelligent behavior*)” (Luger and Stubblefield, 1993)

What is AI?

human vs. rational

thinking

Think like a human
Cognitive Modeling

Think rationally
Logic-based Systems

vs.

acting

Act like a human
Turing Test

Act rationally
Rational Agents

Sumber: S. Russell and P. Norvig. **Artificial Intelligence: A Modern Approach**

Thinking humanly: cognitive modeling

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

- 1960s "cognitive revolution": information-processing psychology, a.k.a. cognitive psychology
- Requires scientific theories of internal activities of the brain
- How to validate? Requires
 - 1) Predicting and testing behavior of human subjects
 - or*
 - 2) Direct identification from neurological data (bottom-up) :
Cognitive Neuroscience
- Both approaches are now distinct from AI
- Caveat: ACT-R & SOAR communities do computational modeling of high level mental functions

We must have some way of determining, how humans think ?

It is in three ways .

1. Through Introspection

2. Through psychological experiment

3. Brain Imaging

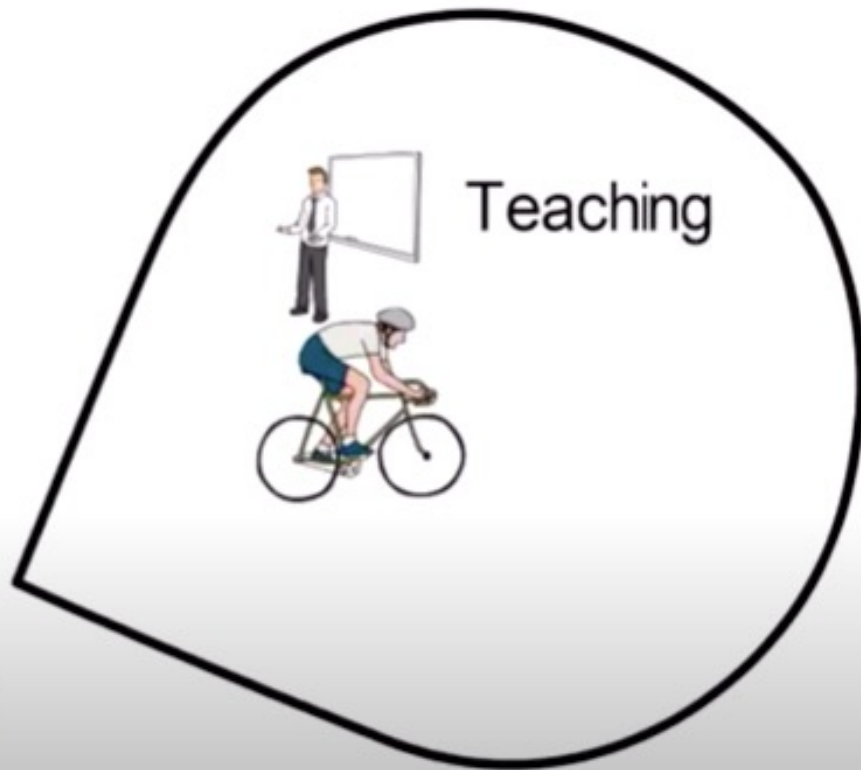


1. Through Introspection

-It is trying to catch our own thoughts as they go by.

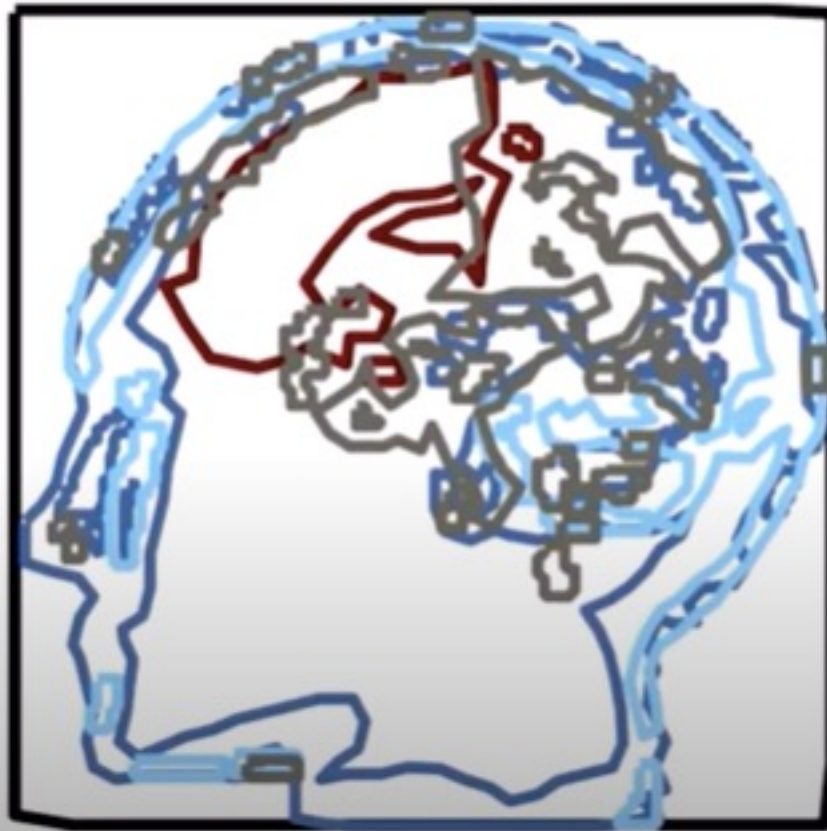


2. Through psychological experiment -It means observing a person in action



3. Brain Imaging

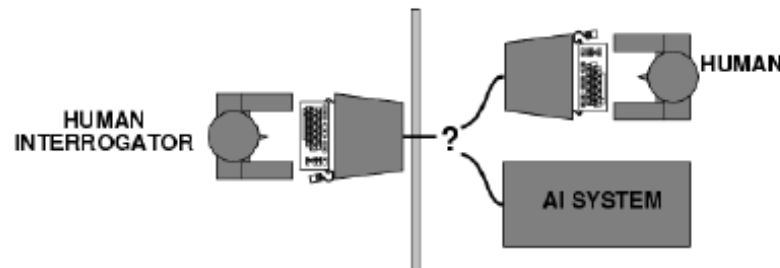
-Observing the brain in action



Acting humanly: Turing Test

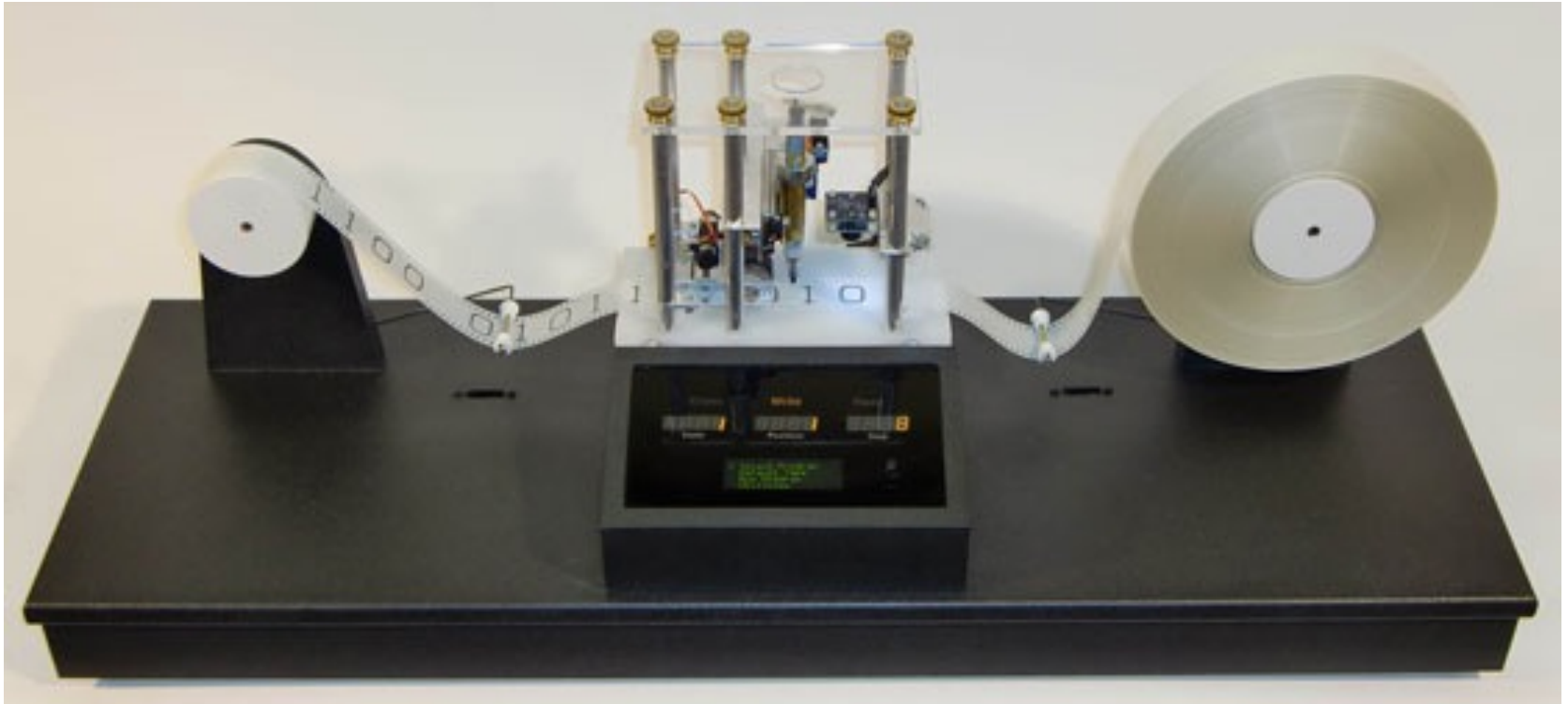
Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

- Turing (1950) "Computing machinery and intelligence":
"Can machines think?" → "Can machines behave intelligently?"
- Operational test for intelligent behavior: the Imitation Game

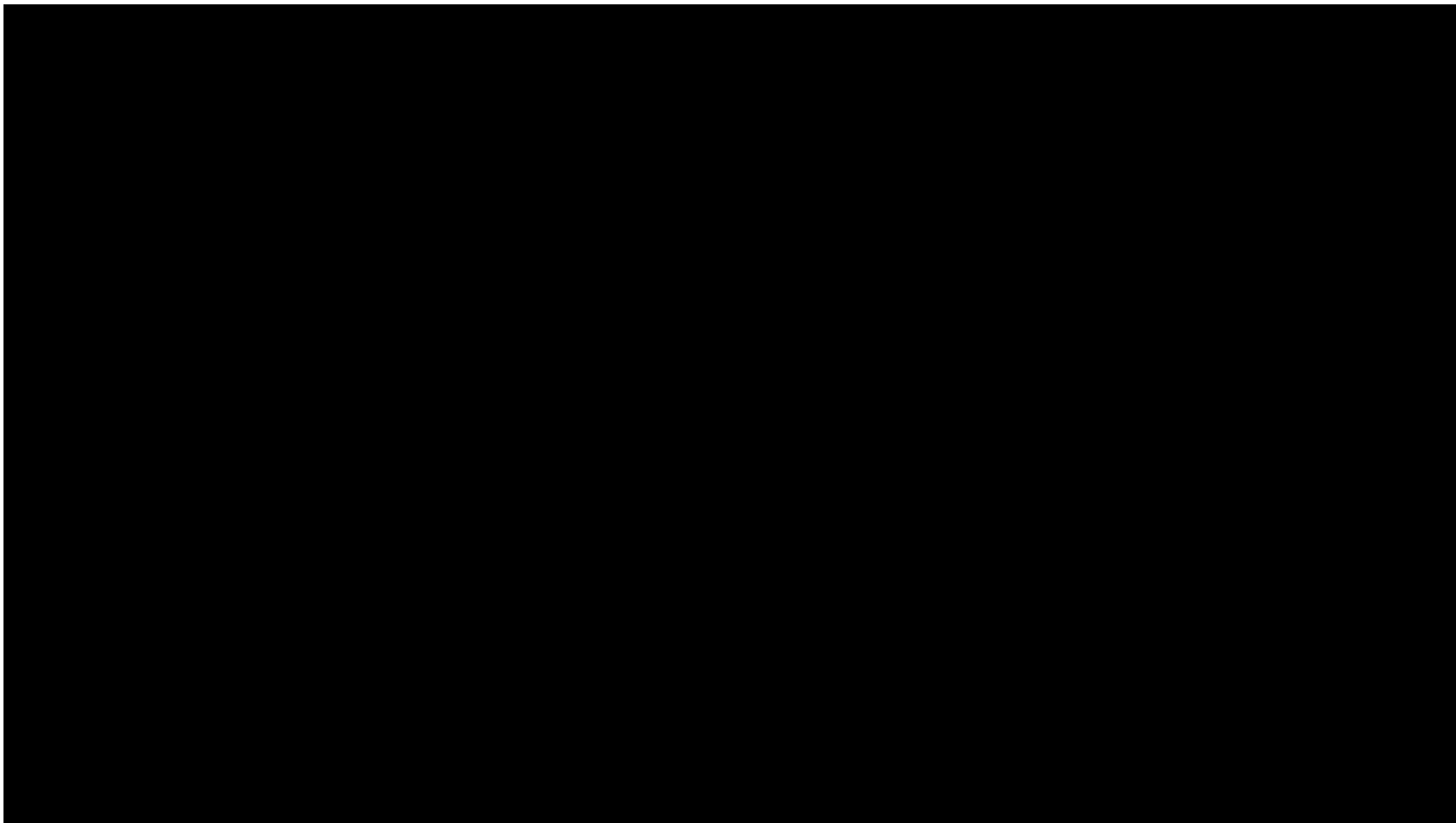


- Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
- Anticipated most major arguments against AI
- Suggested major components of AI: knowledge, reasoning, language understanding, learning

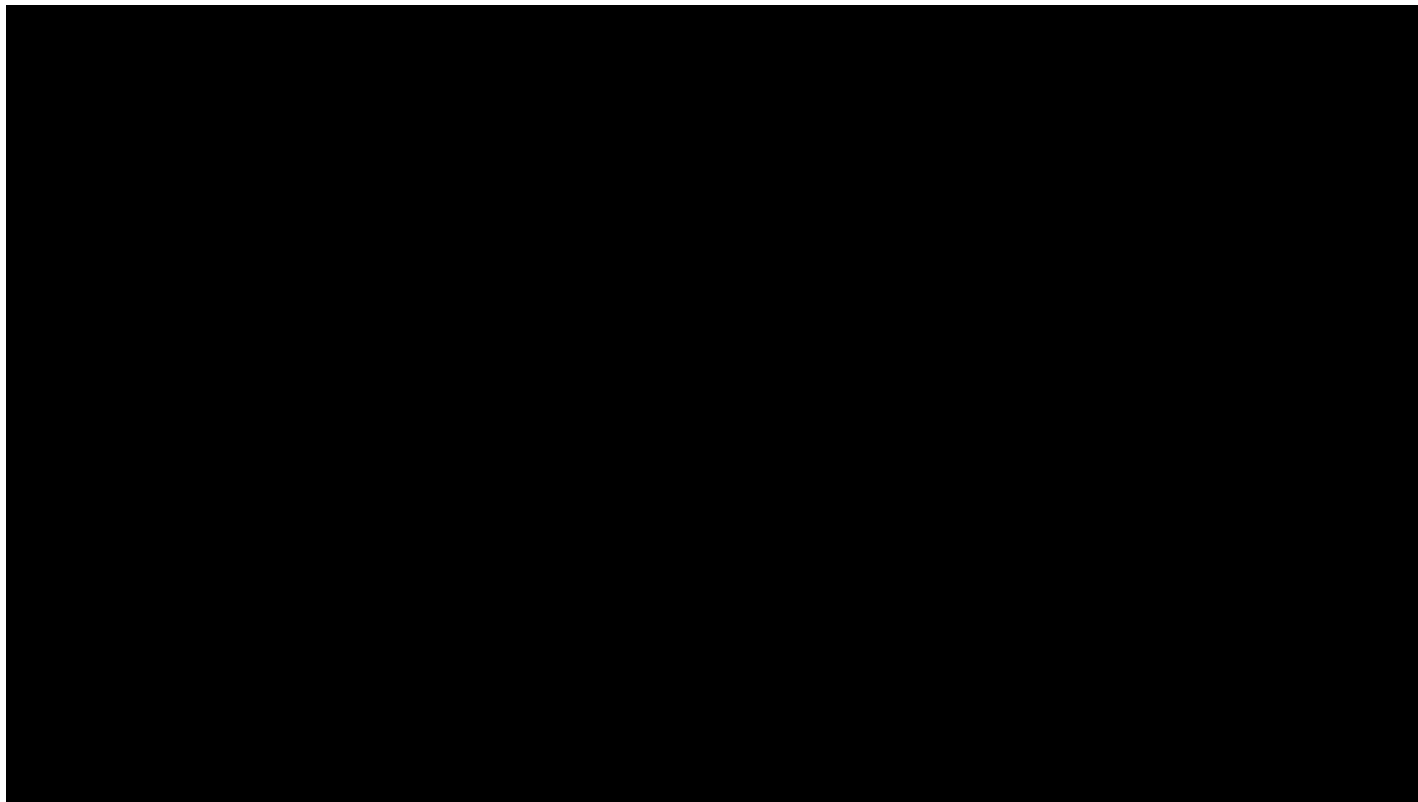
Turing Machine



See Video: <https://www.youtube.com/watch?v=E3keLeMwfHY#t=25>



Mario AI



Thinking rationally: "laws of thought"

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

- Aristotle: what are correct arguments/thought processes?
- Several Greek schools developed various forms of *logic*: *notation* and *rules of derivation* for thoughts; may or may not have proceeded to the idea of mechanization
- Direct line through mathematics and philosophy to modern AI
- **Problems:**
 1. Not all intelligent behavior is mediated by logical deliberation
 2. What is the purpose of thinking? What thoughts should I have?
 3. Ignores the hard problem of perception
 4. All attempts to encode what we know in logic have failed
 5. Most logical inference is *intractable*

2. Thinking Rationally

What is meant of Rationally ?

The ones on the right measure against an ideal performance measure, called rationally .

e.g.

$2 + 2 = 3$ Wrong

$2 + 2 = 4$ Right

THINKING RATIONALLY



How can machine think Rationally ?

If machine is able to develop right logic , then machine can be able to think rationally .

e.g..

Socrates is a man .

All men are mortal .

Therefore Socrates is mortal.

Is Socrates mortal?

Socrates is a man .

All men are mortal .

Therefore Socrates is mortal

Yes.



Acting rationally: rational agents

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

- **Rational** behavior: doing the right thing
- The right thing: that which is *expected to maximize goal achievement, given the available information*
- Doesn't necessarily involve thinking – e.g., blinking reflex – but thinking should be in the service of rational action

Thinking humanly	Thinking rational
Acting humanly	Acting rationally

Rational agents

Rational agent: An *agent* is an entity that perceives and acts

This course is about *effective programming techniques* for designing *rational agents*

Two Approaches to AI

- ***Logical representations:*** BEFORE 1995

- Relations between entities
 - “Mitch’s bicycle is red”
 - (isa B3241 bicycle) (color B3231 red) (owns B3241 P119)
 - (isa P119 person) (name P119 “Mitch”)
- Explicit logical models
- Logical inference, Search
- Chess, Sudoku, computer games, ...

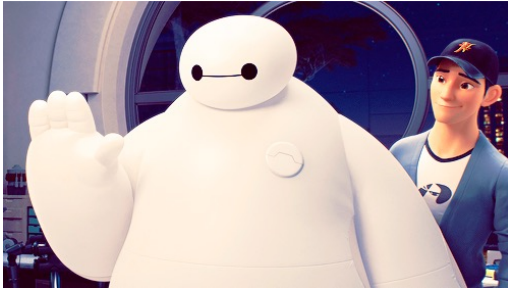
- ***Statistical models:*** SINCE 2000

- Prediction by look-up or by weighted combinations
 - $P(y=\text{bicycle}) = c_0 + c_1 x_1 + c_2 x_2 + c_3 x_3 + \dots$
- Machine Learning, Machine vision, speech recognition, ...



Mengapa
Artificial
Intelligence
penting?

*“AI technology is **important** because it enables human capabilities – understanding, reasoning, planning, communication and perception – to be undertaken by software increasingly effectively, efficiently and at low cost...” - The State of AI 2019: Divergence*



Mengapa Artificial Intelligence Penting?

*Wait 'til my brother sees you!
You're going to help so many
people, buddy. So many!"*

*—Tadashi to Baymax (in a
recording prior to his death)*

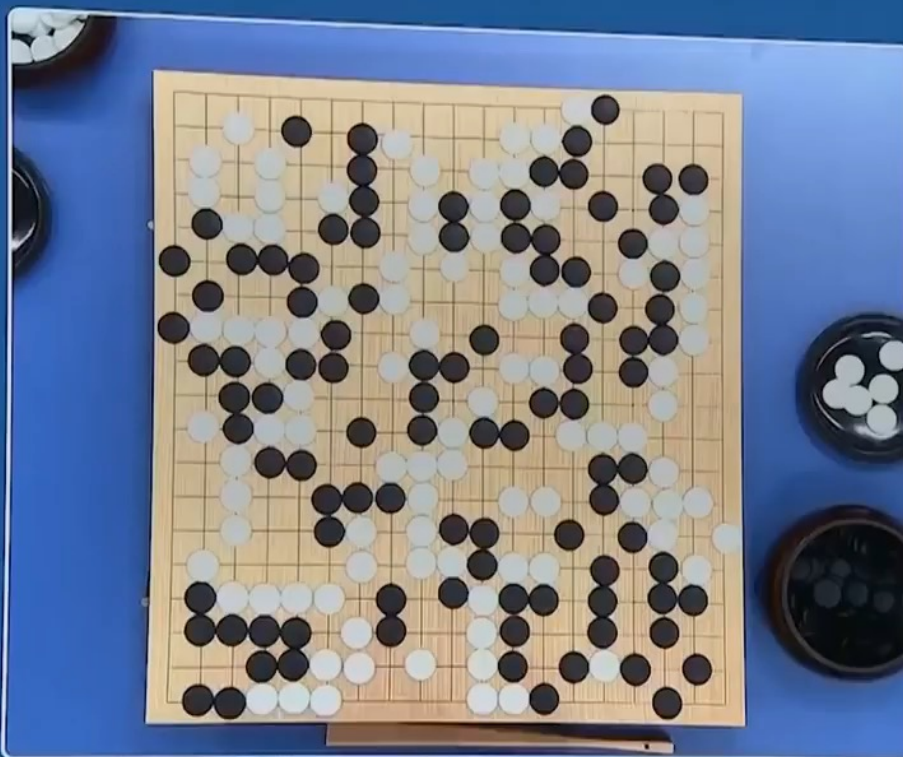




“Garry Kasparov dominated chess until he was beaten by an IBM supercomputer called Deep Blue in 1997...”



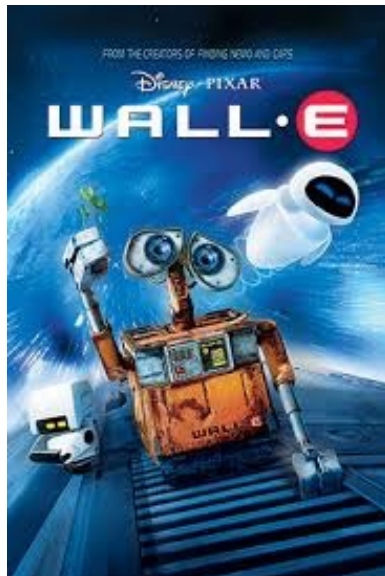
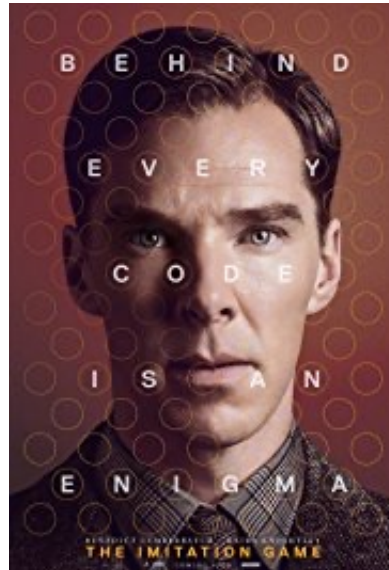
*Google's
AlphaGo AI
defeats world
Go number
one Ke Jie
(May 2017)*



柯洁 KE JIE
00:17:39

ALPHAGO
01:54:35





Film bertema AI

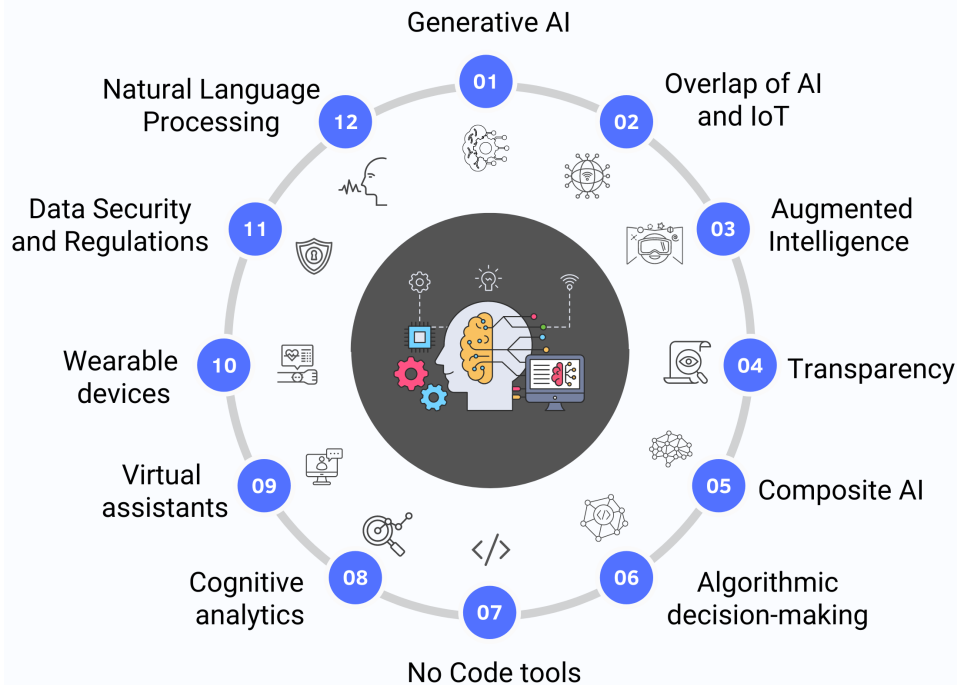
The Terminator

Wall-E

The Imitation Game

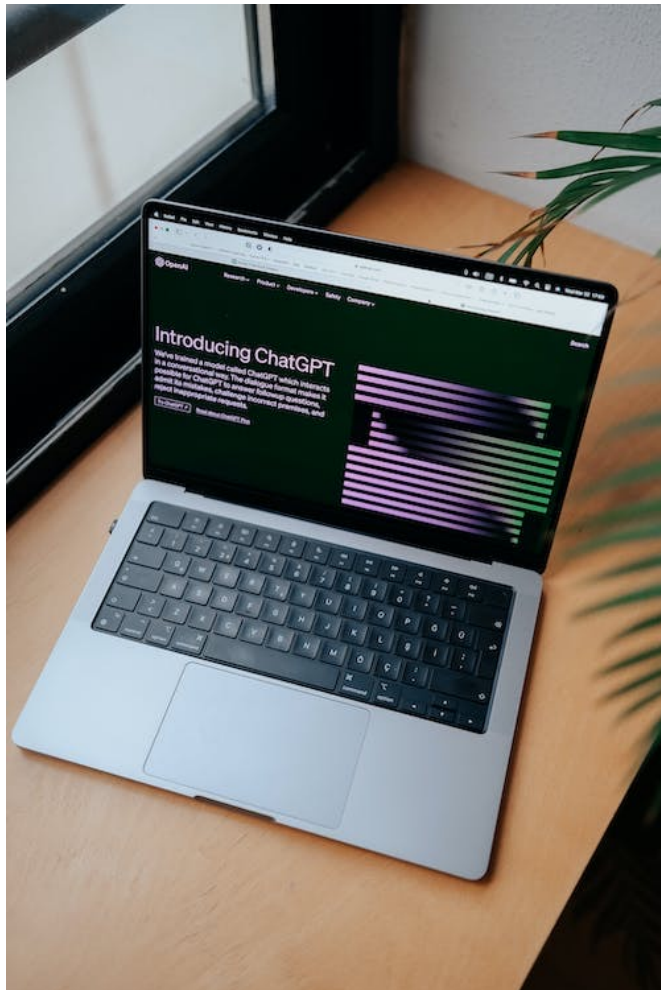
Big Hero 6

2023 Emerging AI and Machine Learning Trends



2023 emerging AI
and Machine
Learning trends

Generative AI



Tobey Maguire instead of Tom Holland in Spider-Man [DeepFake]



Shamook
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Simon Cowell Sings on Stage?! Metaphysic Will Leave You Speechless | AGT 2022



America's Got Talent
25.4M subscribers

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 analytic Steps

OpenAI GPT-3



www.analyticsteps.com





DALL·E 2

DALL·E 2 is a new AI system that can create realistic images and art from a description in natural language.



- JOIN WAITLIST
- ↓ EXPLORE
- ▶ WATCH VIDEO
- 📄 VIEW RESEARCH
- 📷 FOLLOW ON INSTAGRAM



DALL·E 2 can create original, realistic images and art from a text description. It can combine concepts, attributes, and styles.

TEXT DESCRIPTION

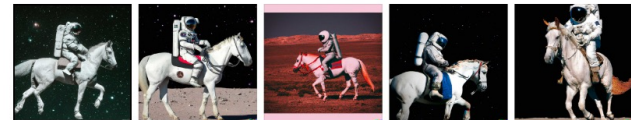
An astronaut Teddy bears A bowl of
soup

riding a horse lounging in a tropical resort
in space playing basketball with cats in
space

in a photorealistic style in the style of Andy
Warhol as a pencil drawing



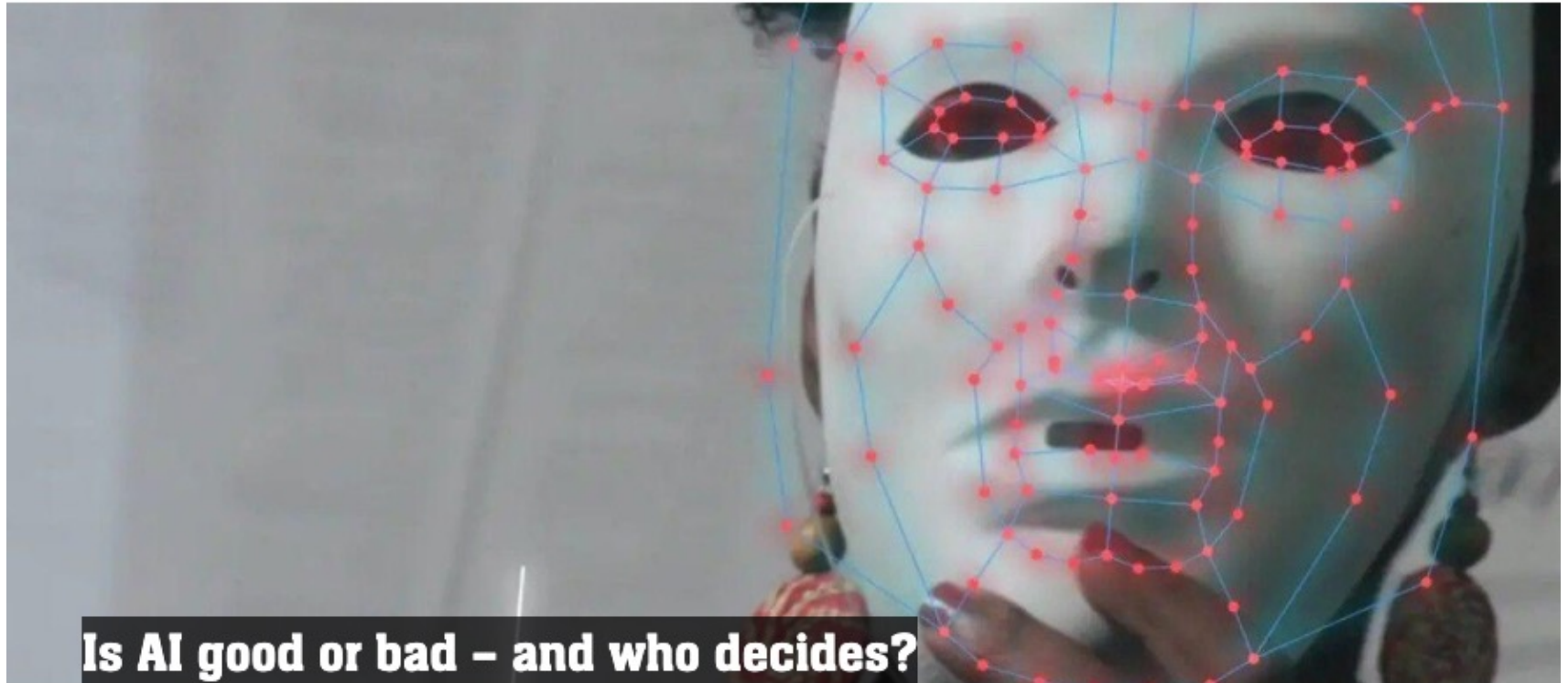
DALL·E 2



What's new in GPT-4?

The technology behind the viral chatbot ChatGPT





Is AI good or bad – and who decides?

By Paul Dempsey

Published Tuesday, August 17, 2021

Deployment of AI is pushing ahead faster than the ethical framework needed to control it.

AI Ethics: What It Is and Why It Matters

In December 2022, the app Lensa AI used artificial intelligence to generate cool, cartoon-looking profile photos from people's regular images. From an ethical standpoint, some people criticized the app for not giving credit or enough money to artists who created the original digital art the AI was trained on [1]. According to The Washington Post, Lensa was being trained on billions of photographs sourced from the internet without consent [2].





Another example is an AI model called ChatGPT, that enables users to interact with it by asking questions. ChatGPT scours the internet for data and answers with a poem, Python code, or a proposal. The ethical dilemma is that people are using ChatGPT to win coding contests or write essays. It raises similar questions to Lensa, but with text rather than images.

These are just two popular examples of AI ethics.

What are AI ethics?

AI ethics refers to the issues that stakeholders (from engineers to government officials) need to consider to ensure artificial intelligence technology is **developed and used responsibly**.

This means taking a safe, secure, humane, and environmentally friendly approach to AI.

AI ethics can cover avoiding bias, ensuring privacy of users and their data, and mitigating environmental risks.

Ethical challenges of AI

AI and bias

If AI doesn't collect data that accurately represents the population, their decisions might be susceptible to bias. In 2018, Amazon was under fire for its AI recruiting tool that downgraded resumes that featured "women" (such as "Women's International Business Society"). In essence, the AI tool discriminated against women and caused legal risk for the tech giant.

AI and privacy

As mentioned earlier with the Lensa AI example, AI relies on data pulled from internet searches, social media photos and comments, online purchases, and more. While this helps to personalize the customer experience, there are questions around the apparent lack of true consent for these companies to access our personal information.

AI and the environment

Some AI models are large and require significant amounts of energy to train on data. While research is being done to devise methods for energy-efficient AI, more could be done to incorporate environmental ethical concerns into AI-related policies.

How to create more ethical AI

Creating more ethical AI requires the evaluation of policy, education, and technology. Regulatory frameworks can ensure that technologies benefit society rather than harm it.

Globally, governments are beginning to enforce policies for ethical AI, including how companies should deal with legal issues if bias or other harm arises.

Stakeholders in AI Ethics

Each of these actors play an important role in ensuring less bias and risk for AI technologies.

Academics: Researchers and professors are responsible for developing theory-based statistics, research, and ideas that can support governments, corporations, and non-profit organizations.

Government: Agencies and committees within a government can help facilitate AI ethics in a nation. A good example of this is the [Preparing for the Future of Artificial Intelligence](#) report that was developed by the National Science and Technology Council (NSTC) in 2016, which outlines AI and its relationship to public outreach, regulation, governance, economy, and security.

Intergovernmental entities: Entities like the United Nations and the World Bank are responsible for raising awareness and drafting agreements for AI ethics globally. For example, UNESCO's 193 member states adopted the first ever global agreement on the [Ethics of AI](#) in November 2021 to promote human rights and dignity.

Non-profit organizations: Non-profit organizations like Black in AI and Queer in AI help diverse groups gain representation within AI technology. The Future of Life Institute created 23 guidelines that are now the Asilomar AI Principles, which outline specific risks, challenges, outcomes for AI technologies.

Private companies: Executives at Google, Meta, and other tech companies, as well as banking, consulting, health care, and any other industry that uses AI technology, are responsible for creating ethics teams and codes of conduct. This often creates a standard for companies to follow suit.

**“ I FAIL TO SEE HOW
YOU FAIL TO SEE THAT
THIS IS
AWESOME ! ”**

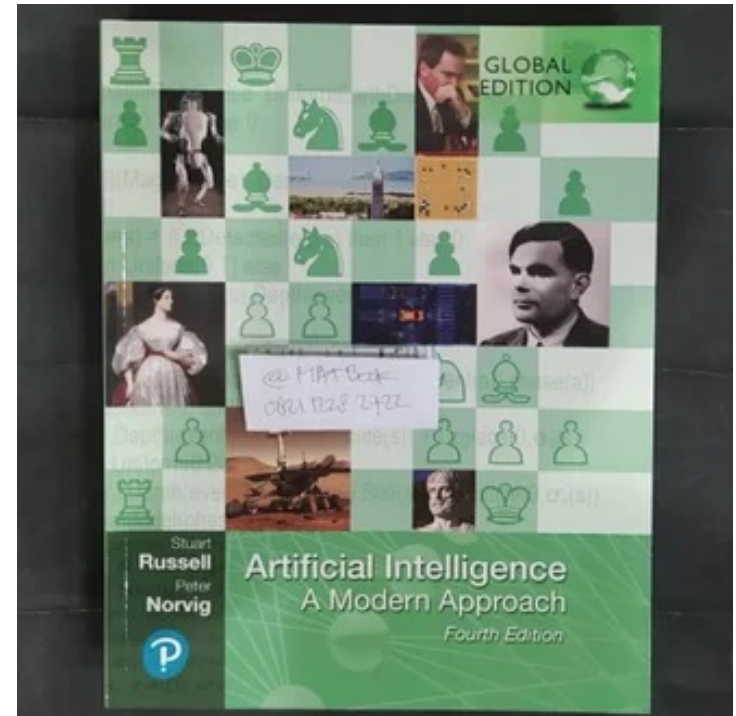
-BIG HERO 6



Tugas

Baca Chapter 1: Introduction

Buku: S. Russell and P. Norvig.
Artificial Intelligence: A Modern Approach. Prentice Hall, 2009, Third Edition (U.S.)



Next Week:
Agent &
Environment
(Chap 2)

Referensi

S. Russell and P. Norvig. Artificial Intelligence: A Modern Approach. Prentice Hall, 2009, Third Edition (U.S.)

<http://ai.berkeley.edu/home.html>

Hem Chandra Live – Youtube Channel

And other various sources